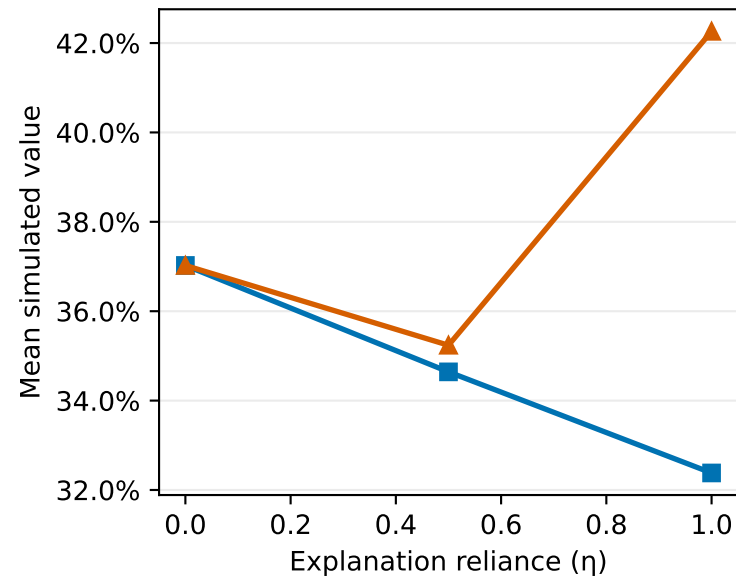


Theory-only sweep: Explanation reliance (η)

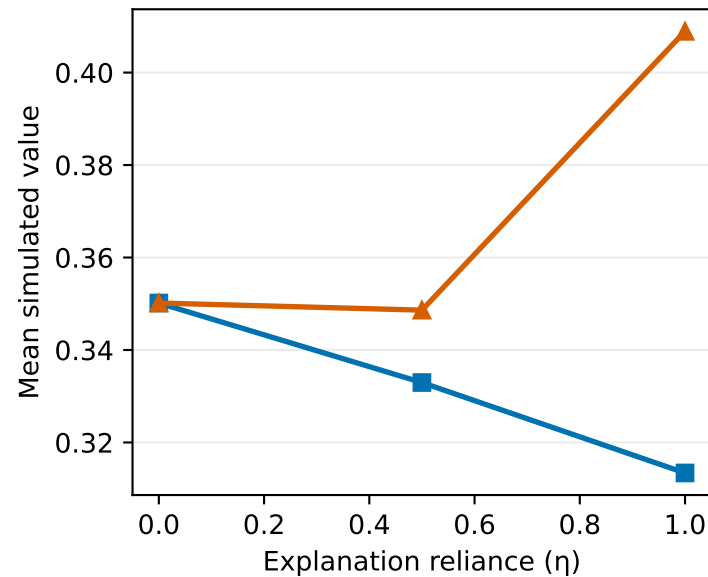
Feature-contribution family · K=2 · other parameters uniformly marginalized

■ Feature attribution ▲ Counterfactual example

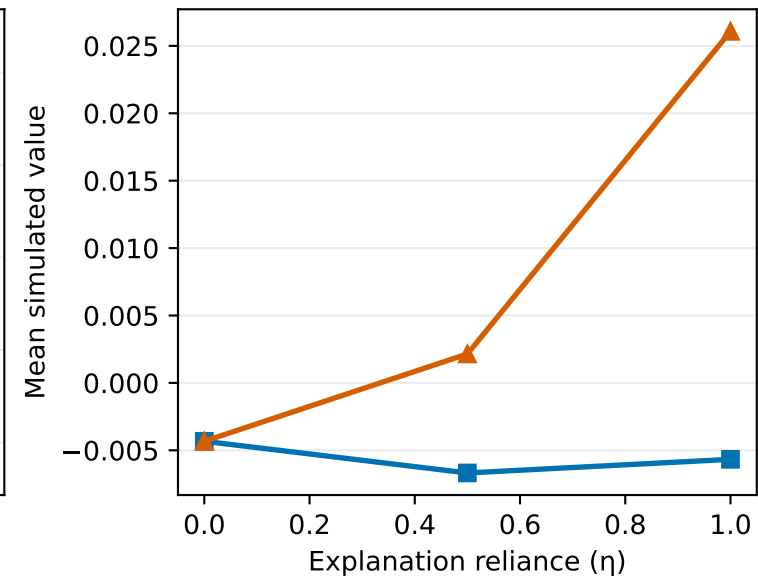
Counterfactual success rate



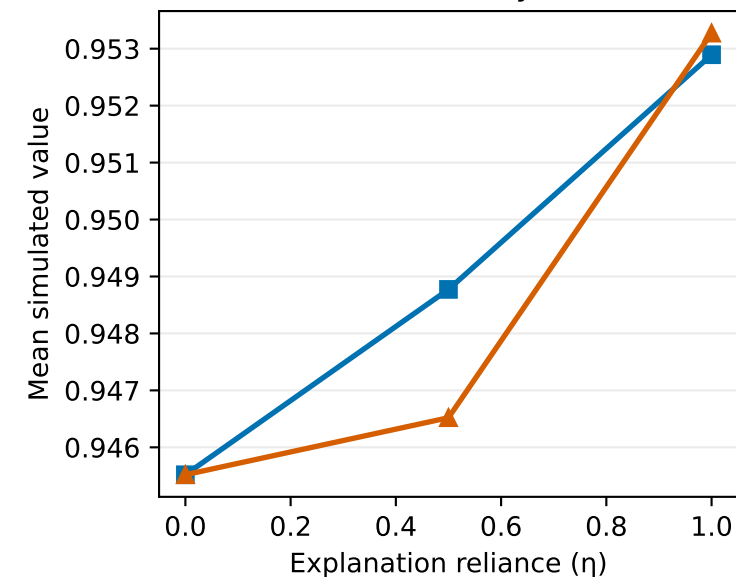
Target-confidence gain



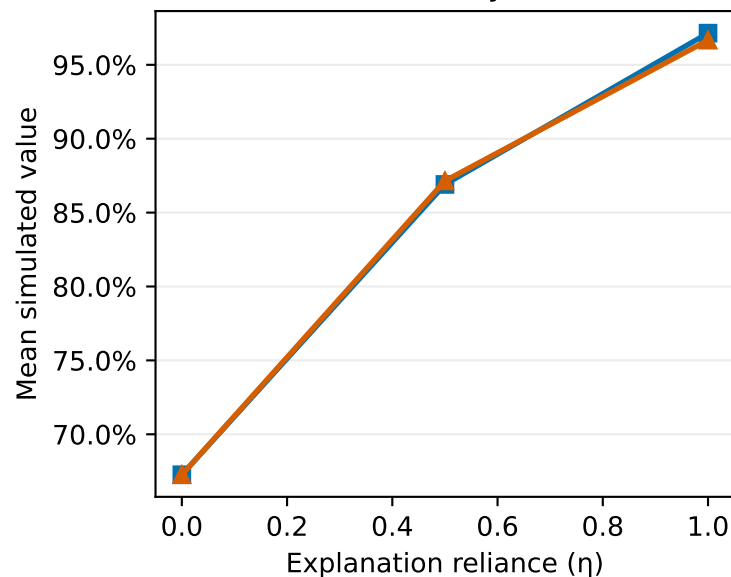
Boundary-distance improvement



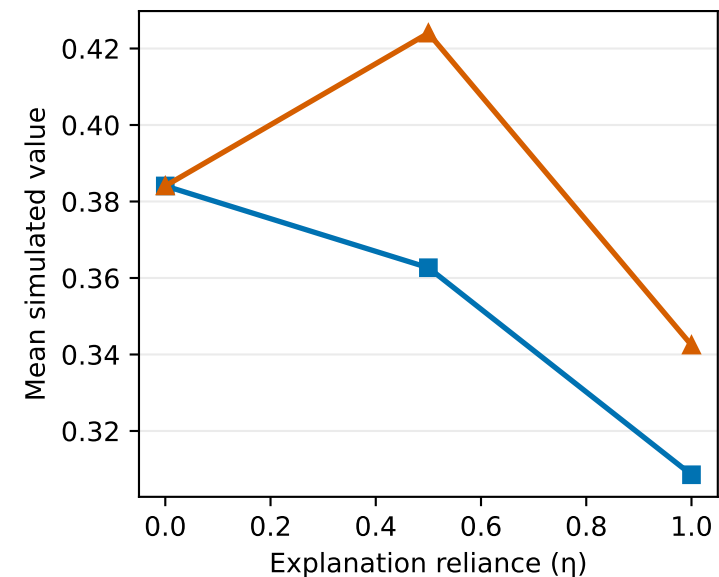
Plausibility



Actionability rate



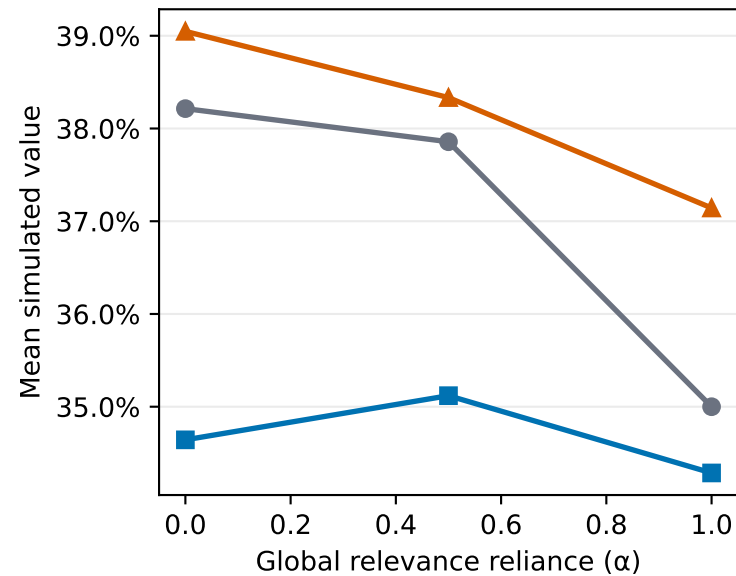
Normalized edit distance (L1)



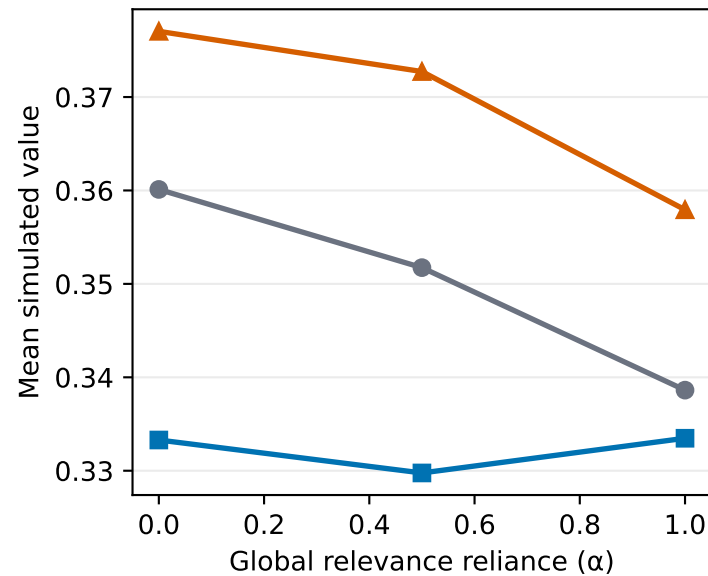
Theory-only sweep: Global relevance reliance (α)
Feature-contribution family · K=2 · other parameters uniformly marginalized

—●— No XAI —■— Feature attribution —▲— Counterfactual example

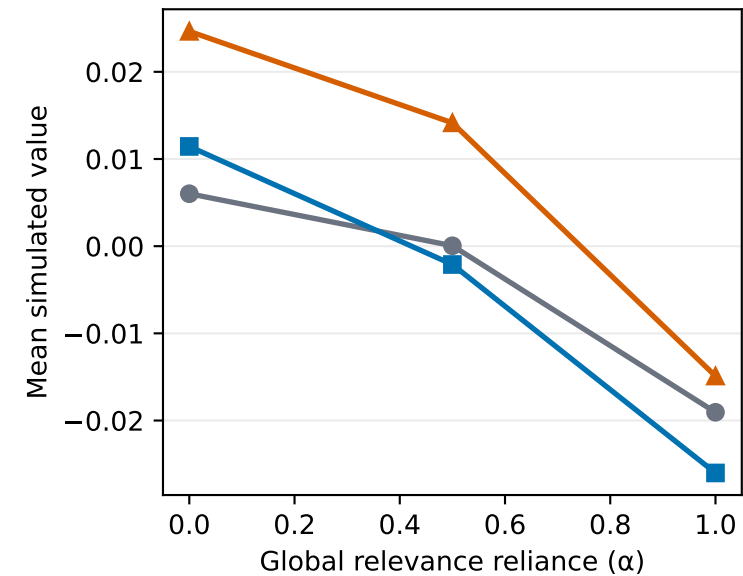
Counterfactual success rate



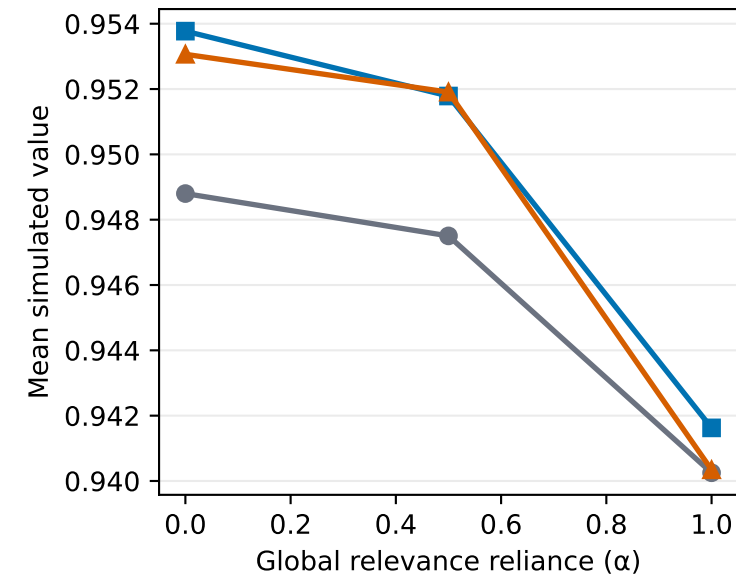
Target-confidence gain



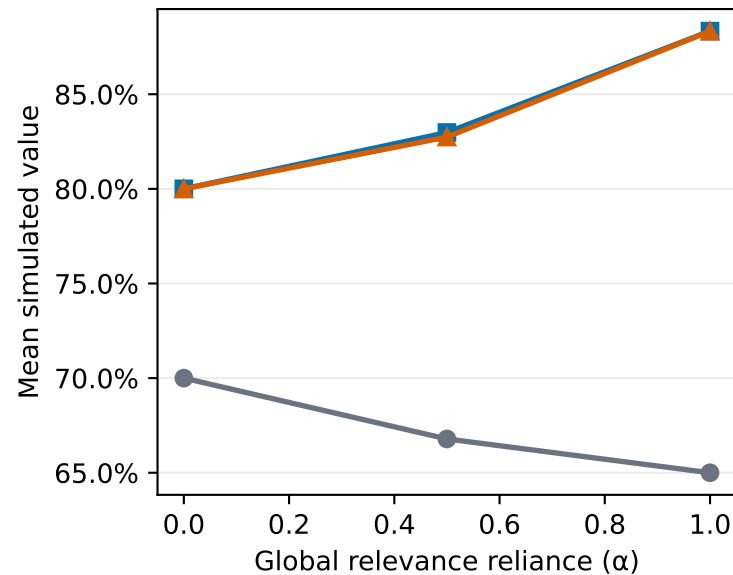
Boundary-distance improvement



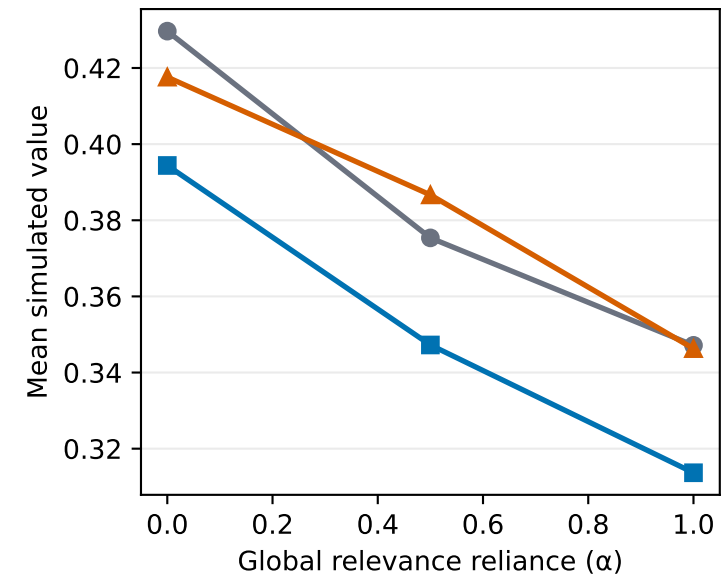
Plausibility



Actionability rate



Normalized edit distance (L1)

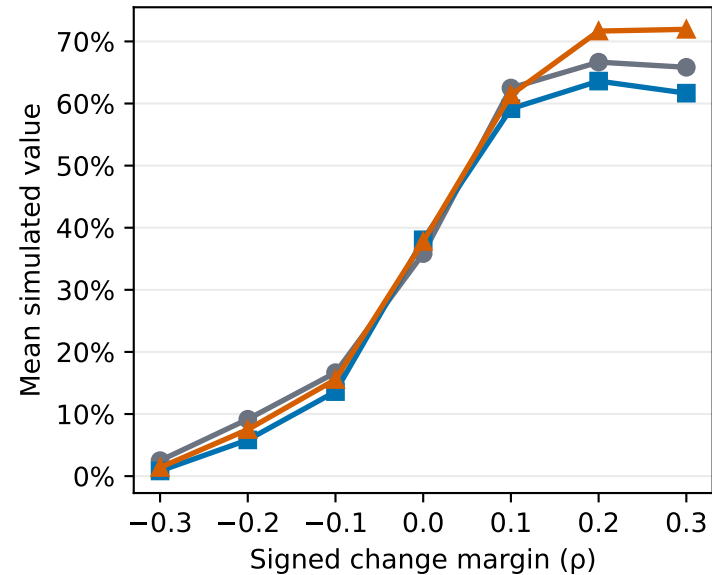


Theory-only sweep: Signed change margin (ρ)

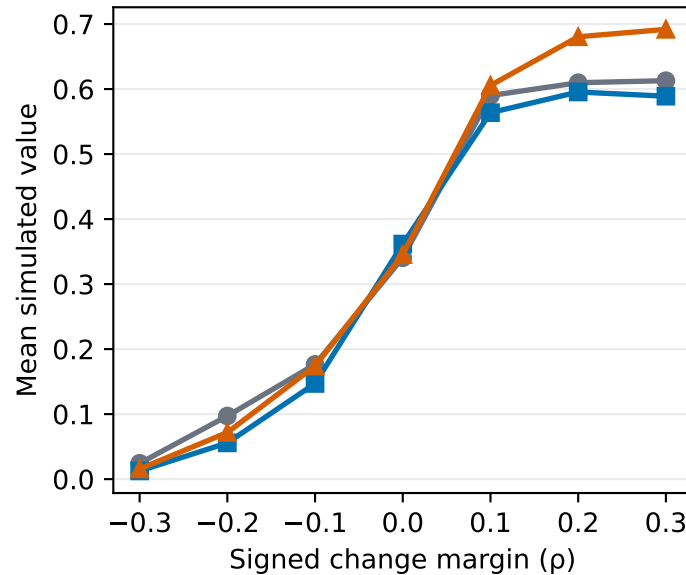
Feature-contribution family · K=2 · other parameters uniformly marginalized

—●— No XAI —■— Feature attribution —▲— Counterfactual example

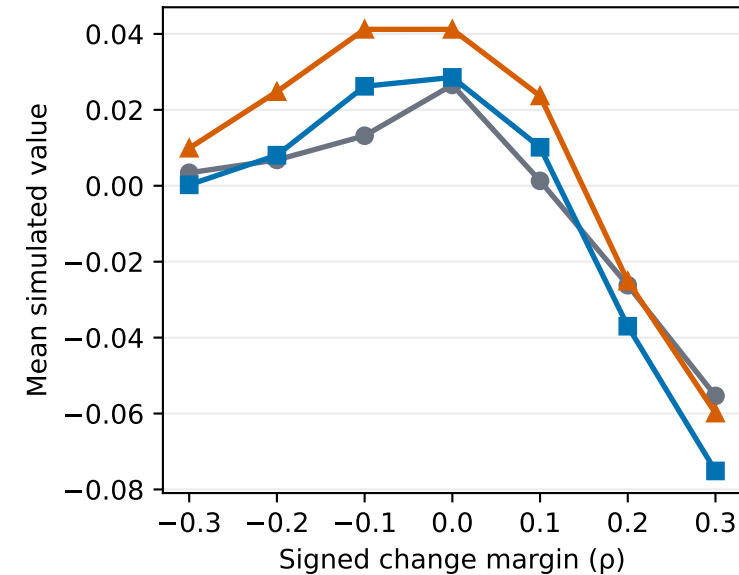
Counterfactual success rate



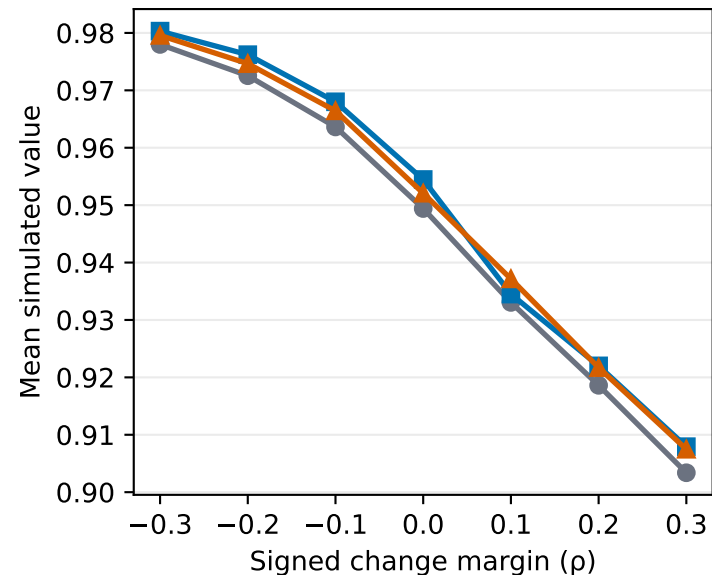
Target-confidence gain



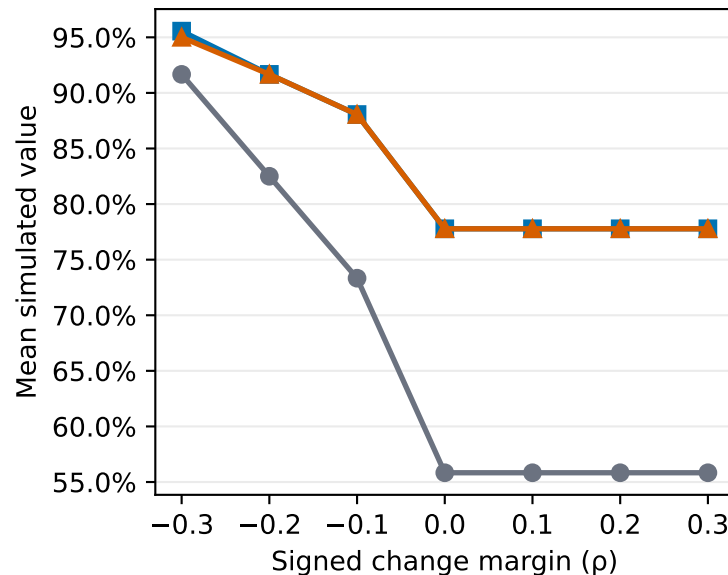
Boundary-distance improvement



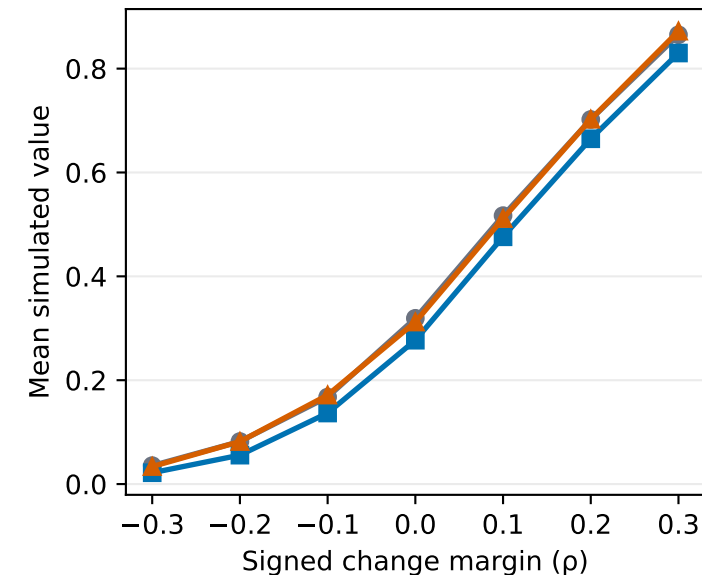
Plausibility



Actionability rate



Normalized edit distance (L1)

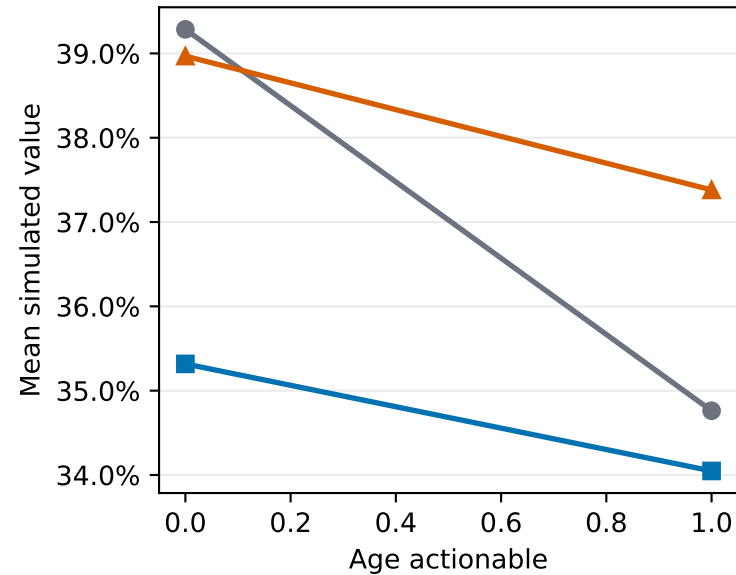


Theory-only sweep: Age actionable

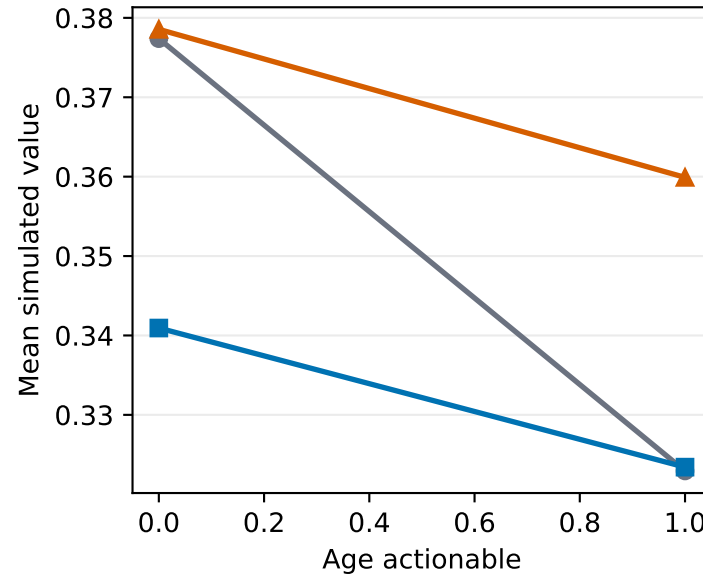
Feature-contribution family · K=2 · other parameters uniformly marginalized

Legend: No XAI (grey circle), Feature attribution (blue square), Counterfactual example (orange triangle)

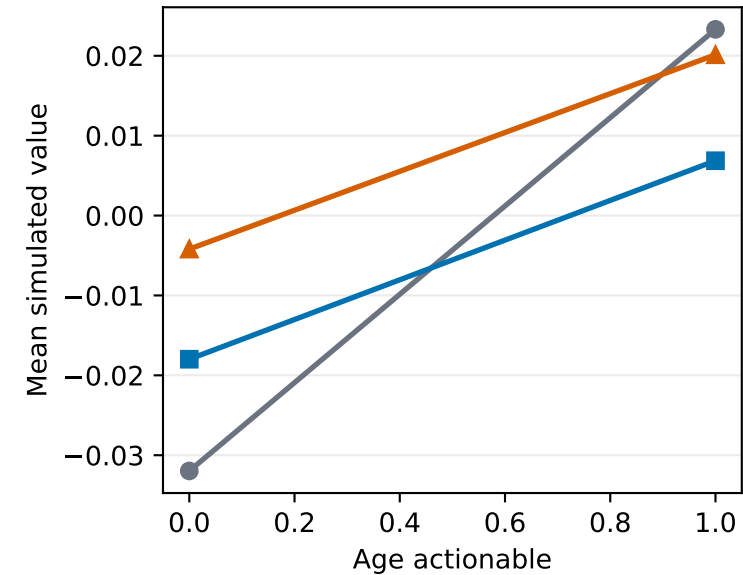
Counterfactual success rate



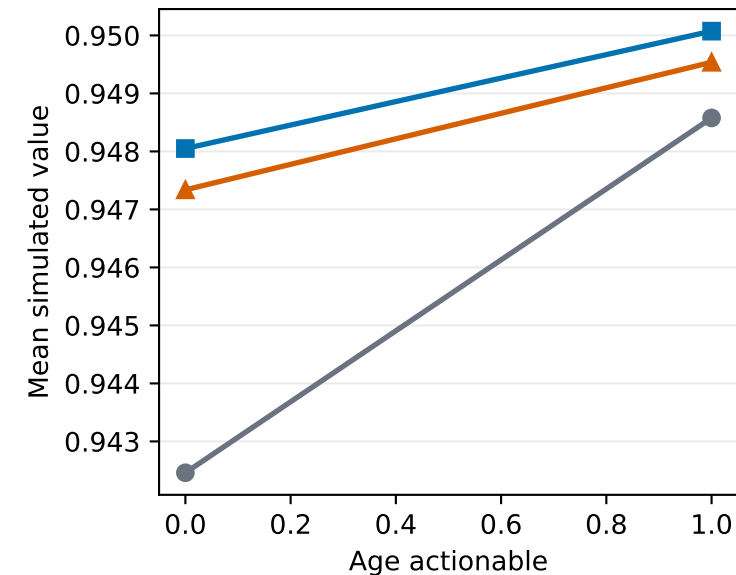
Target-confidence gain



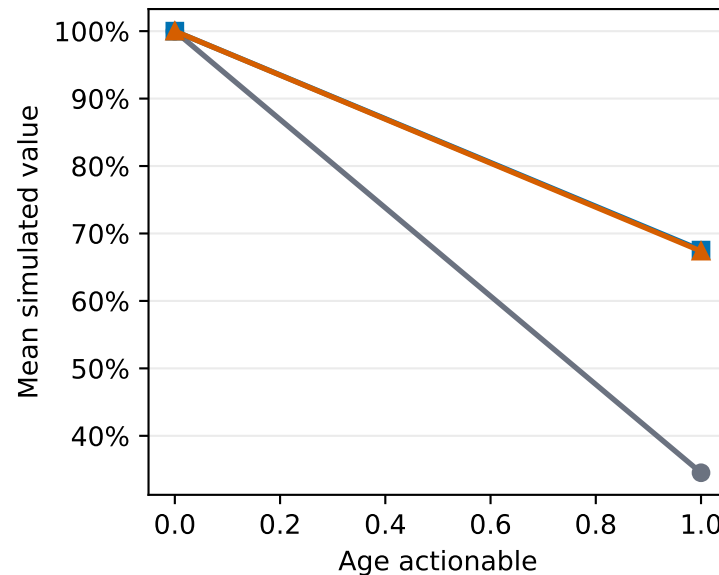
Boundary-distance improvement



Plausibility



Actionability rate



Normalized edit distance (L1)

